

PROJECT:

ProtectR- Autonomous Edge-AI Wearable for Real-Time Women's Safety & Emergency

OBJECTIVES:

- Detect threats automatically without requiring user intervention.
- Trigger instant emergency alerts without internet.
- Provide real-time location for rapid rescue.
- Capture tamper-proof evidence during dangerous situations.
- Protect victim identity using blockchain technology.
- Enable faster response and automated FIR processing.

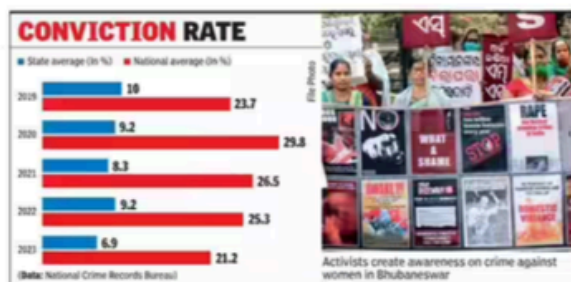
PROBLEM STATEMENT:

Women facing harassment, stalking, or sudden threats may be unable to access their phones or activate SOS systems during emergencies. **Existing safety solutions rely on smartphones, internet connectivity, and manual activation, creating critical gaps in emergency protection.** Therefore, there is a need for a discreet, intelligent, and reliable hands-free safety system that can automatically detect threats, send instant alerts, share location, and preserve evidence without depending on internet connectivity.

Conviction rate in crimes against women rose to 41% in 1st 6 months: Data

Debabrata Mohapatra / Dec 11, 2025, 04:11 IST

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Bhubaneswar: Conviction rate in 'crimes against women' in the state surged to 41% in the first half of this year, an all-time high for Odisha police, data revealed. According to NCRB data, conviction rates in these cases remained abysmally low for years, consistently hovering between 8% and 10% during 2019-2023.

INTRODUCTION:

Women's safety requires intelligent protection that can respond instantly when conventional safety methods fail. **This project presents a smart, discreet, and hands-free wearable safety system designed to continuously monitor movement and voice cues for potential emergencies.** Using edge AI, motion sensors, and audio-based threat detection, the device identifies abnormal gestures, distress keywords, and threatening situations without requiring manual activation. A multi-modal threat assessment combines motion, audio, and location data to determine emergency severity. **When danger is detected, the system automatically triggers alerts, shares real-time location, and securely captures evidence through encrypted storage.** Offline GSM and LoRa communication ensures emergency connectivity even when internet access is unavailable, providing rapid and reliable protection.

SOFTWARE REQUIRED:

Software / Tool	Purpose
Arduino IDE	Develop and upload wearable device firmware
TensorFlow / Keras	Develop and train AI detection models
Edge Impulse	Build and deploy TinyML models
Google Maps	Real-time location tracking and visualization
Firebase	Store and manage emergency alert data
GitHub	Version control and project collaboration
Blockchain Platform	Secure identity and evidence records
Figma	Design system interface and project prototype
VS Code	Develop and manage project software
Serial Monitor	Monitor and debug sensor communication

HARDWARE REQUIRED:

Hardware Component	Purpose
MEMS Microphone	Detect distress keywords and threatening sounds
GPS Module	Capture the user's real-time location
3-Axis Accelerometer	Detect abnormal movement and sudden motion
Gyroscope	Identify unusual gestures and body movements
Low-Power MCU	Control sensors and process safety events
Light Sensor	Detect surrounding light and night conditions
GSM Module	Send emergency alerts through SMS
LoRa Module	Enable communication without internet connectivity
Wearable Enclosure	Conceal the device as an everyday accessory

SYSTEM WORKFLOW:

- **Continuous Real-Time Monitoring** – Accelerometer, gyroscope, and microphone continuously monitor movement and voice patterns.
- **Abnormal Activity Detection** – Edge AI automatically identifies unusual motion, distress keywords, or threatening sounds.
- **Multi-Modal Threat Analysis** – Motion, audio, and location data are intelligently combined to calculate threat level.
- **Emergency Threat Verification** – The system accurately classifies the event as normal, medium, or high threat.
- **Automatic Evidence Capture** – During high-risk situations, audio, motion logs, location, and timestamps are securely recorded.
- **Secure Data Encryption** – Evidence and user information are securely protected using encryption and blockchain technology.
- **Offline Emergency Alert Transmission** – GSM/LoRa sends emergency alerts and location without requiring internet connectivity.
- **Immediate Emergency Response** – Trusted contacts, police authorities, and safety volunteers receive alerts for quick assistance.
- **Automated FIR Processing** – In extreme abuse cases, the system prepares an FIR after discreet user verification.

WORKFLOW DIAGRAM:



Fig1: System workflow illustrating how the wearable detects threats, captures evidence, sends offline alerts, shares location, and enables rapid emergency response.

HARDWARE IMPLEMENTATION :

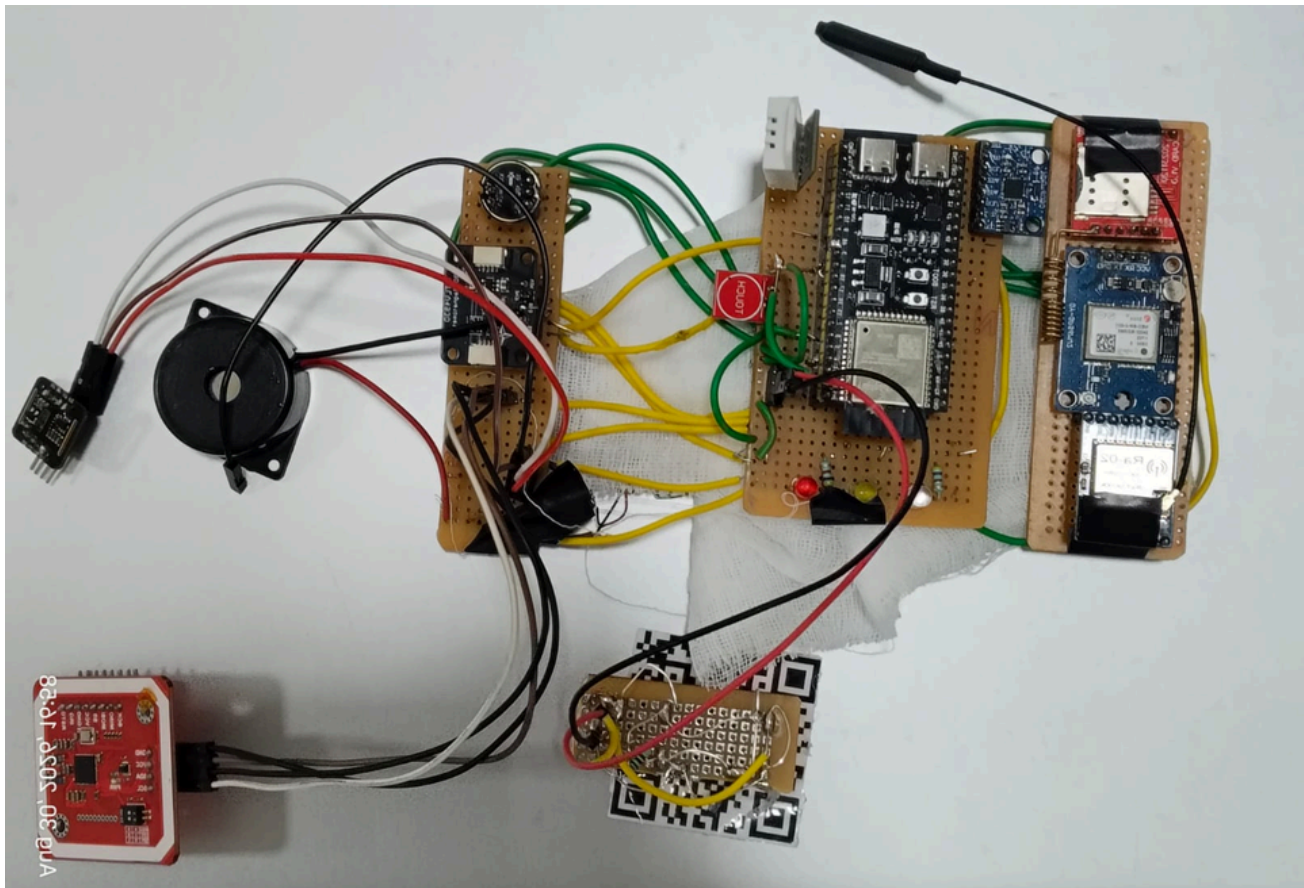


Fig2: Wearable safety prototype for intelligent threat detection and emergency response.

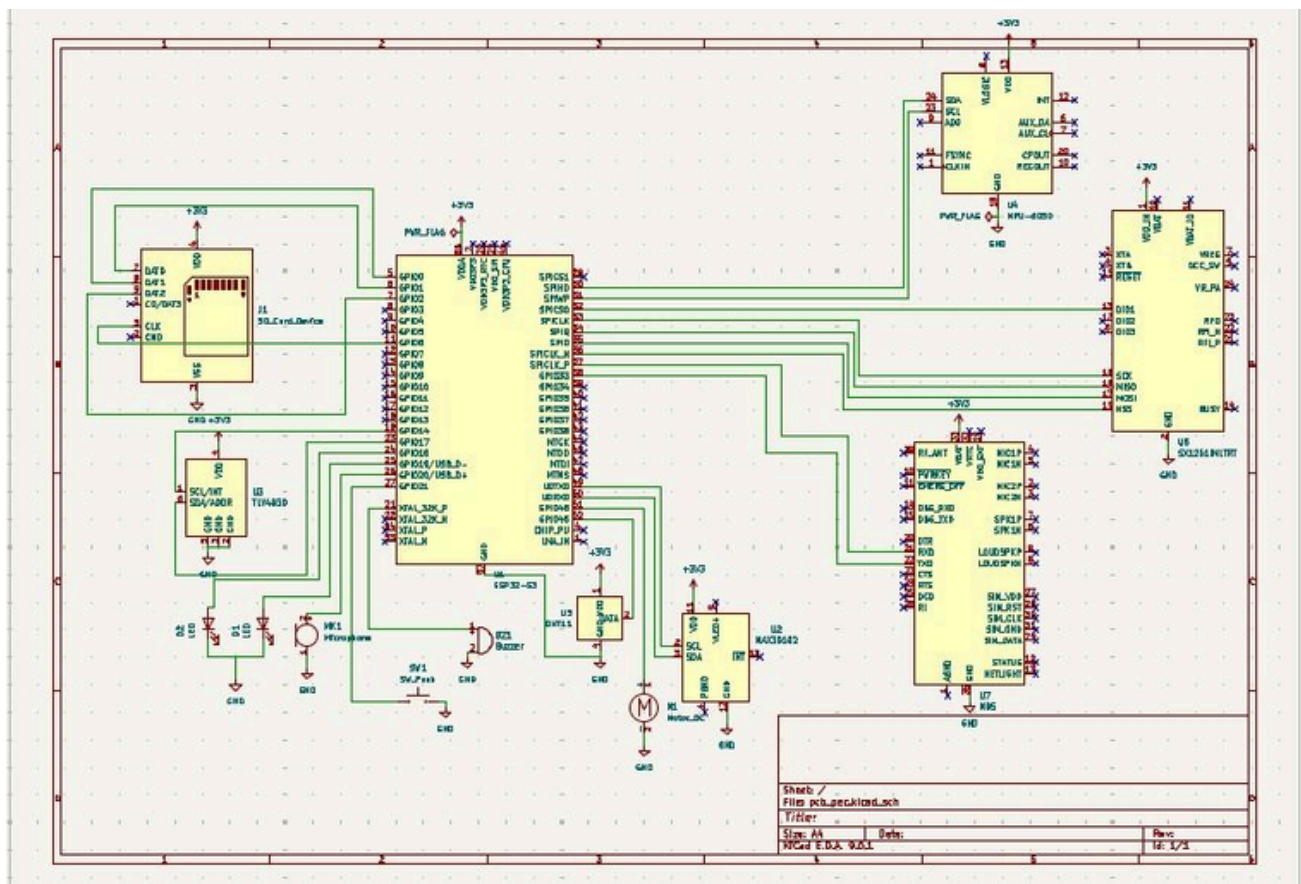


Fig3: KiCad circuit diagram illustrating the complete hardware connections for the women's safety system.

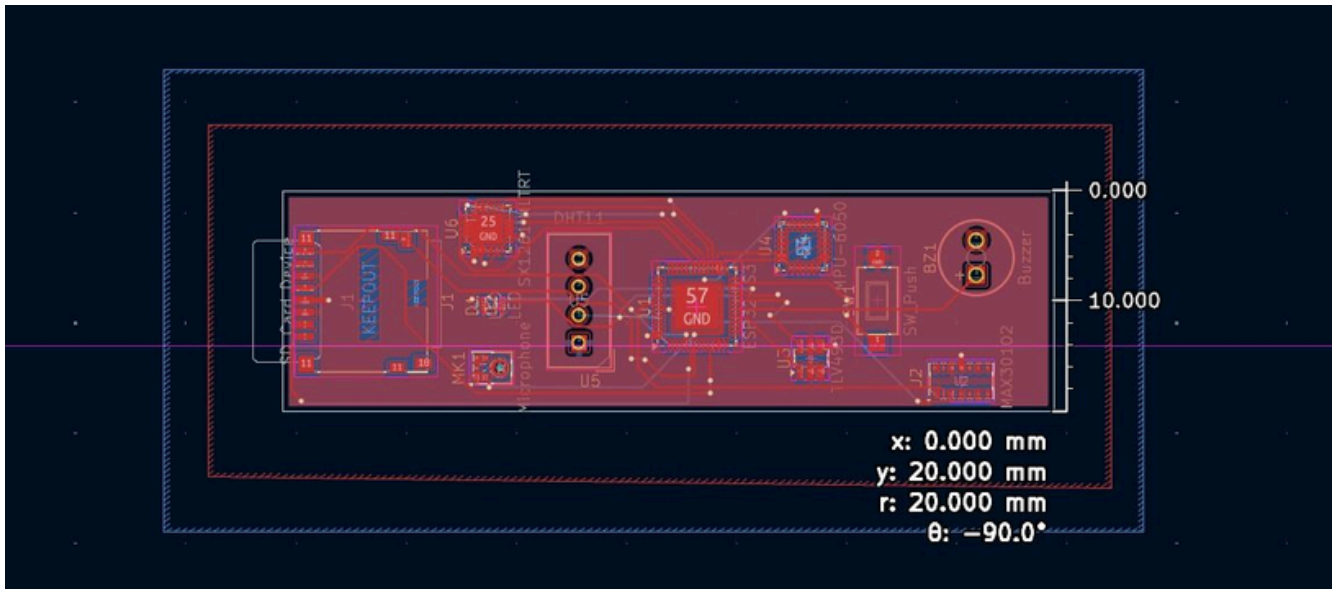
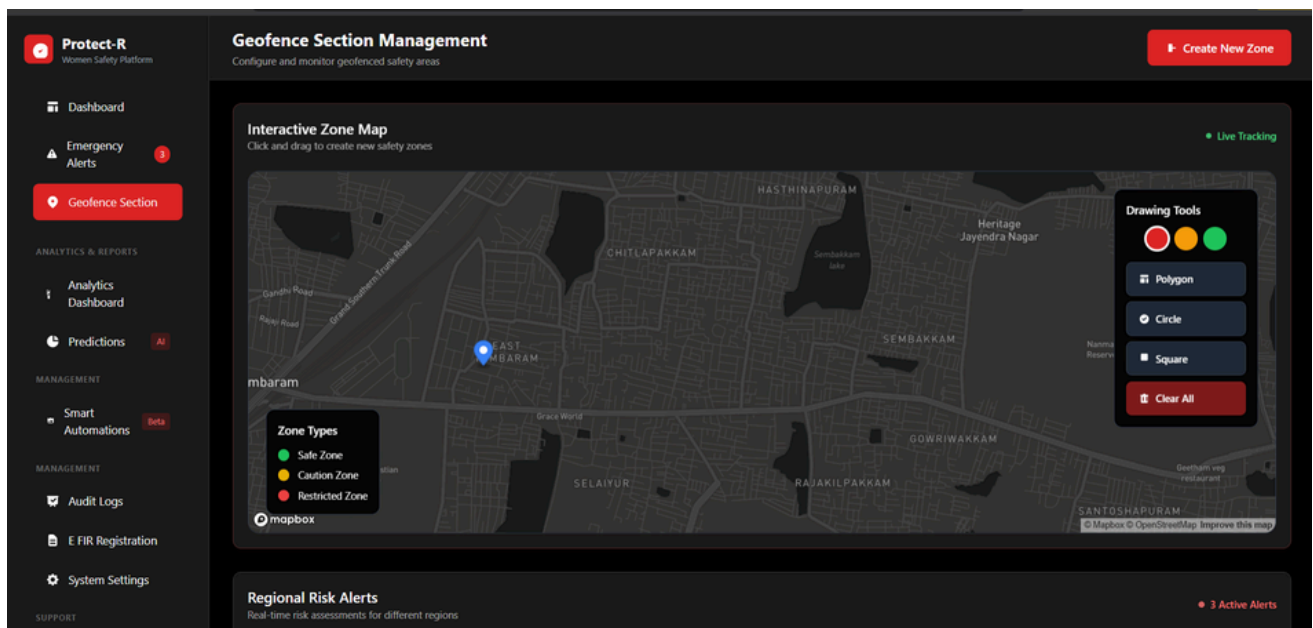
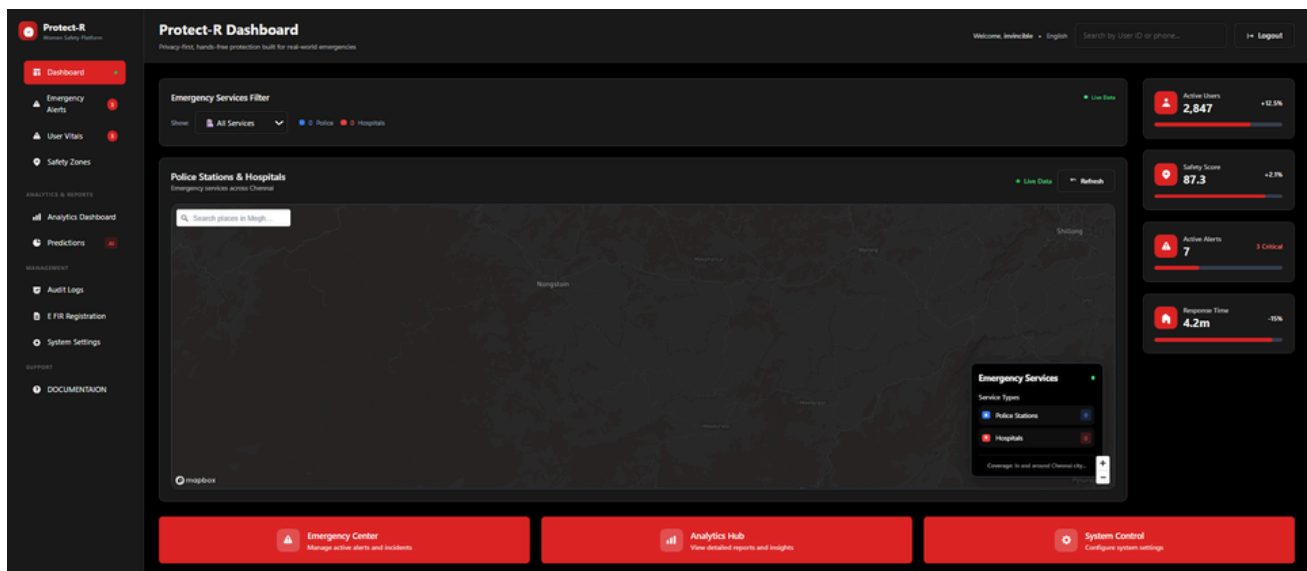


Fig3: KiCad PCB design illustrating the optimized board layout and component connections for the women's safety system.

SOFTWARE IMPLEMENTATION :



 SafeGuard
Tourist Safety

Dashboard

Emergency Alerts

Geofence Section

E FIR Registration

ANALYTICS & REPORTS

Analytics Dashboard

Predictions AI

MANAGEMENT

Audit Logs

Settings

E FIR Registration

First Information Report (E-FIR) — Government of Tamil Nadu

Download E FIR

FIRST INFORMATION REPORT (E-FIR)

Government of Tamil Nadu - Tamil Nadu Police Department

District

CHENNAI

Police Station

Virtual Cyber Station

Year

2026

FIR No.

Auto-generated on Submit

Date of Report

Auto-filled on Submit

Date of Incident

dd-mm-yyyy

Time of Incident

--:--

1. Complainant Details

Name

Enter complainant's full name

 Protect-R
Women Safety Platform

Dashboard

Emergency Alerts 3

Audit Logs

E FIR Registration

System Settings

SUPPORT

Help Center

Women Safety Analytics

Monitor safety metrics and user activity patterns

Last 30 days

Export Report

Key Performance Indicators

Total Safety Score

87.3

+2.1%

Out of 100

Active Tourists

2,847

+12.5%

Currently tracked

Incidents Today

7

-1

5 resolved, 2 pending

Response Time

4.2m

-15%

Average response

Analytics Overview

Safety Trends

Safety Score

Tourist Count

Jan Feb Mar Apr May Jun

Incident Types

Last 7 days

Medical

Lost

Weather

Security

Equipment

Women User Page:

The women user portal provides personal safety monitoring, including live vitals, real-time location, geofencing, emergency alerts, nearby police stations and hospitals, and access to e-FIR services.

Police Page:

The police portal enables authorities to monitor active users and emergency situations, view live locations and vitals, receive alerts, track geofencing violations, access safety heatmaps, and respond quickly to reported incidents.

Authority Page:

The authority portal provides a centralized safety management dashboard with incident analytics, emergency trends, heatmaps, e-FIR monitoring, user statistics, location insights, and overall system performance for informed decision-making.

PROTOTYPE COST:

Component	Estimated Cost (₹)
MEMS Microphone	₹150
GPS Module	₹300
GSM Module	₹400
LoRa Module	₹350
Accelerometer + Gyroscope	₹200
Light Sensor	₹50
Low-Power MCU	₹300
PCB & Fabrication	₹500
Battery & Power Supply	₹250
Wearable Enclosure	₹200
Total Prototype Cost	₹2,700

BUSINESS ANALYTICS:



KEY FEATURES OF PROTECT-R:

- **Hands-Free Threat Detection** - Detects abnormal movements, distress keywords, and threatening situations automatically.
- **Offline Emergency Communication** - Uses GSM and LoRa to send alerts even without internet.
- **Multi-Modal AI Detection** - Combines motion, audio, and location data for improved threat assessment.
- **Automatic Evidence Capture** - Records audio, motion logs, location, and timestamps during high-risk situations.
- **Blockchain-Based Privacy** - Protects user identity and maintains tamper-proof evidence.
- **Automated E-FIR Support** - Prepares an FIR for extreme abuse cases after discreet user verification.
- **Camouflaged Wearable Design** - Designed as everyday accessories, making the system discreet and difficult to disable.

FEASIBILITIES OF PROTECT-R:

- **Readily Available Hardware** – Uses practical components such as GPS, microphone, motion sensors, GSM, and LoRa modules.
- **Edge AI Processing** – Threat detection runs locally, reducing dependence on cloud connectivity.
- **Low Computational Blockchain Use** – Blockchain is limited to identity and evidence hashing for efficient operation.
- **Offline Communication Capability** – GSM/LoRa enables emergency communication during network or internet failures.
- **Scalable Implementation** – The system can support B2C, B2B, and B2G deployment models.
- **Practical Wearable Architecture** – Low-power MCU and compact components support continuous monitoring.

ONE LINE USP:

An intelligent, hands-free, offline safety system that detects threats, protects identity, preserves evidence, and enables rapid emergency response.

CONCLUSION:

The proposed Women's Safety System provides an intelligent, proactive, and reliable approach to personal safety by combining AI-based threat detection, wearable sensing, GPS tracking, GSM/LoRa communication, secure evidence management, and real-time emergency response. The system is designed to function even in challenging network conditions while reducing dependence on manual intervention.

By integrating hands-free threat detection, offline communication, geofencing, location tracking, tamper-resistant evidence, and e-FIR support, the project creates a complete safety ecosystem connecting women, police, authorities, and emergency services. Its scalable B2C, B2B, and B2G model further supports practical deployment and long-term social impact.